

Module 09: Granularity of Data Items and Hierarchy

AKTU BCS501 | Unit 5: Concurrency Control Techniques

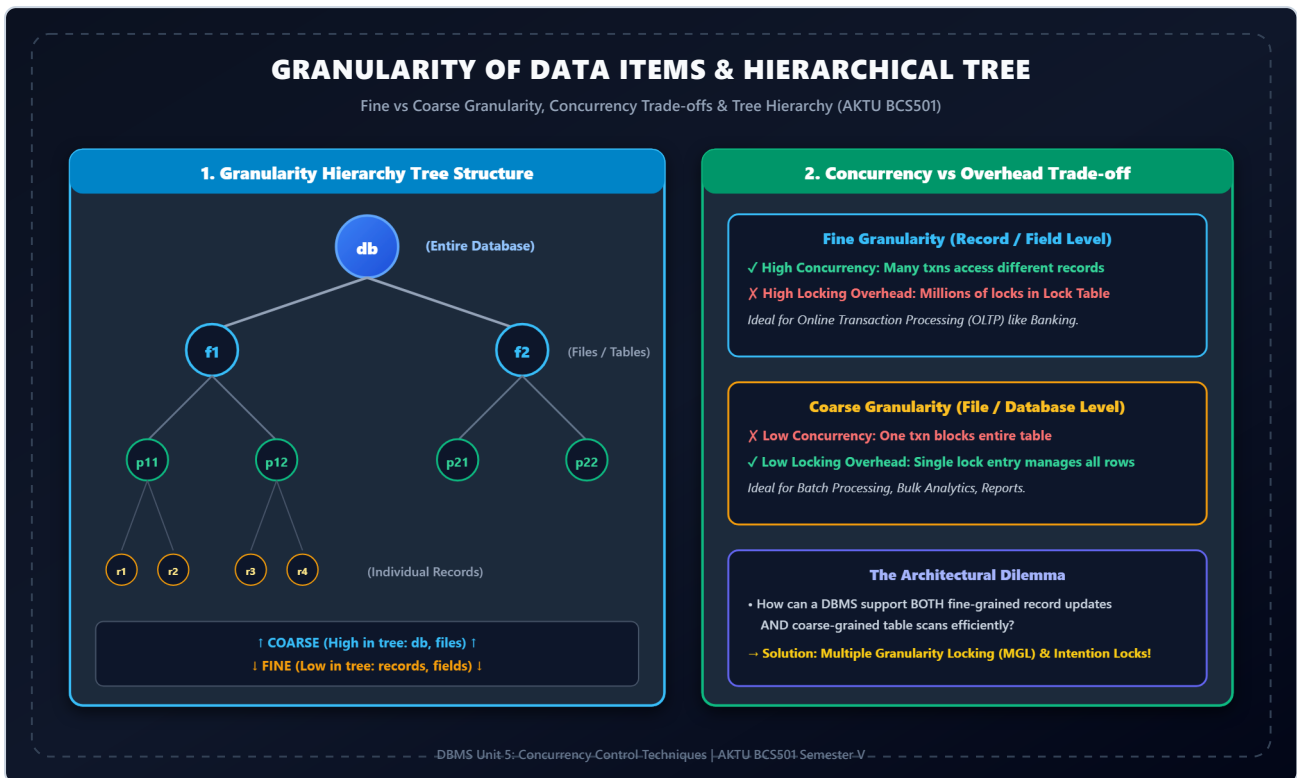


Figure 9.1: Granularity Tree Hierarchy & Concurrency vs Overhead Trade-offs

1. Granularity Definition

Granularity refers to the size of data items chosen for locking (database, files, pages, records, fields).

2. Concurrency vs Overhead Trade-off

Fine Granularity (Record): Maximizes concurrent access, but incurs heavy lock table overhead.

Coarse Granularity (File/Database): Minimal locking overhead, but restricts concurrency.